Adaptation Strategies

EES 4760/5760

Agent-Based and Individual-Based Computational Modeling

J. Magnolia Gilligan

Class #13: Monday, October 12, 2026

Live web page: https://ees4760.jmgilligan.org/slides/class_13
PDF: https://ees4760.jmgilligan.org/slides/class_13/EES_4760_5760_class_13_slides.pdf

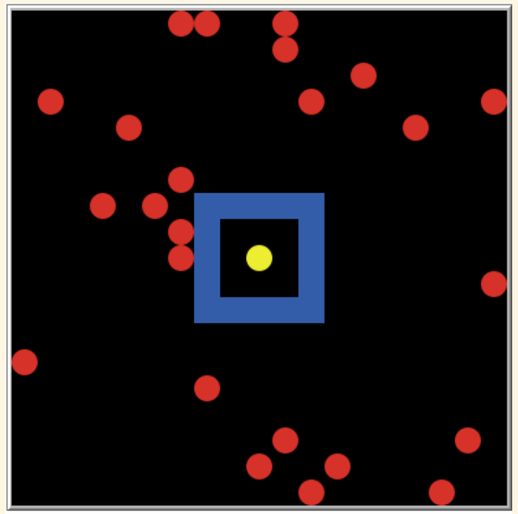
Getting Started

- Sit with your team partners
- Download model:
  - https://ees4760.jmgilligan.org/models/class_13/business-investor.nlogox
  - or go to Downloads page,
    - select “Adaptation Strategies” page,
    - and download Business Investor model.

Subsetting

Subsetting

- Open the BusinessInvestor model in NetLogo
- Click `setup`
- Turn all the turtles red and the patches black
- Turn `turtle 5` yellow and move it to `patch 0 0`
- Ask `turtle 5` to identify all the patches that are exactly 2 patches away from the turtle's patch (not a 2-patch radius from `turtle 5`)



Hints:

- There are many ways to do this. Let's look at a way to do this with the `neighbors` primitive.
- Hints:
  - Use `member?` primitive (`member? <agent> <agent-set>`)
  - Use `patch-set` primitive to turn an list of many patch-sets into a single patch-set
- Suggestion:
  1. Start by turning all neighbor patches (patches exactly 1 patch away) blue
  2. Next turn all patches within 2 patches blue
  3. Now turn all patches black again
  4. Now turn all patches within a 2-patch distance blue *except* the turtle's patch
  5. Now turn all patches black again
  6. Now turn all patches within a 2-patch distance blue *except* the turtle's patch and the patches 1 patch away.

A solution

```
ask turtle 5 [
  ask (patch-set [neighbors] of [neighbors] of self) with
    [not member? self [(patch-set neighbors patch-here)] of myself]
  [
  set pcolor blue
  ]
]
```

- What does `self` refer to in `(patch-set [neighbors] of [neighbors] of self)`?
  - `self` refers to `turtle 5`
  - `ask turtle 5 [ ... ]` puts the `[...]` in the context of `turtle 5`, so `self` refers to `turtle 5`
- What does `self` refer to in `not member? self [(patch-set neighbors patch-here)] of myself`?
  - `self` refers to the various patches in the `patch-set`: `(patch-set [neighbors] of [neighbors] of self)`
  - `x with [...]`, where `x` is an `agent-set` evaluates `[...]` for each of the agents (patches, turtles, links) in `x`, so `self` in the `[...]` refers, in turn, to each patch in the `patch-set`
- What does `myself` refer to in `not member? self [(patch-set neighbors patch-here)] of myself`?
  - `myself` refers to `turtle 5`
  - `myself` refers to the agent doing the asking

Self vs. Myself

```
to test-self-myself
  ask turtle 5
  [
    ask turtle 7
    [
      print (word "first self = " self)
      print (word "first myself = " myself)
      ask turtle 2
      [
        print (word "second self = " self)
        print (word "second myself = " myself)
      ]
    ]
  ]
end
```

```
observer> test-self-myself

first self = (turtle 7)
first myself = (turtle 5)
second self = (turtle 2)
second myself = (turtle 7)
```

- `self` refers to the agent *being asked*.
- `myself` refers to the agent *doing the asking*.
- First: `turtle 5` is asking `turtle 7` to do something.
  - `self` is `turtle 7`, `myself` is `turtle 5`
- Second: `turtle 7` is asking `turtle 2` to do something.
  - `self` is `turtle 2`, `myself` is `turtle 7`

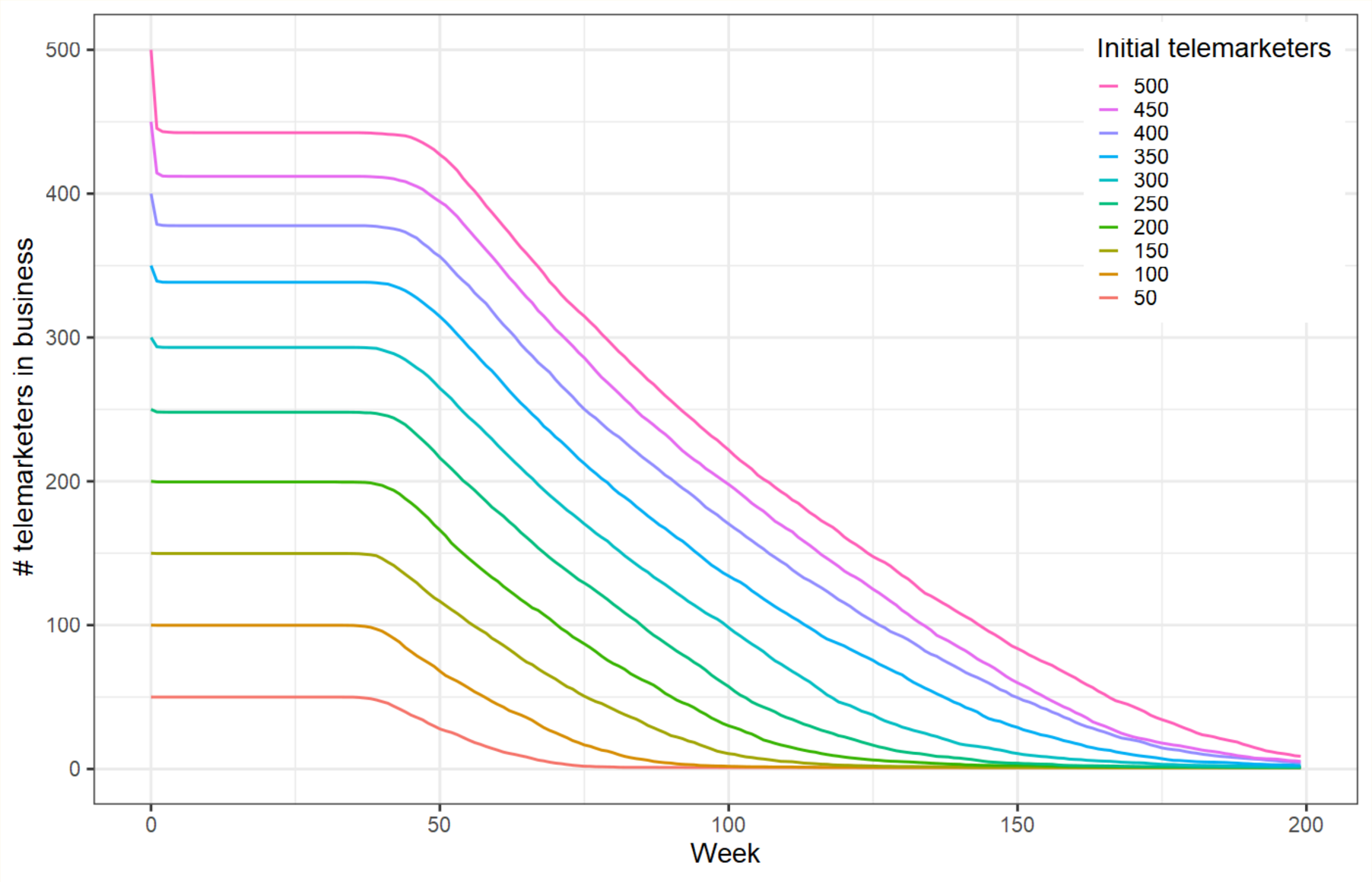
Telemarketer Model

Telemarketer Model

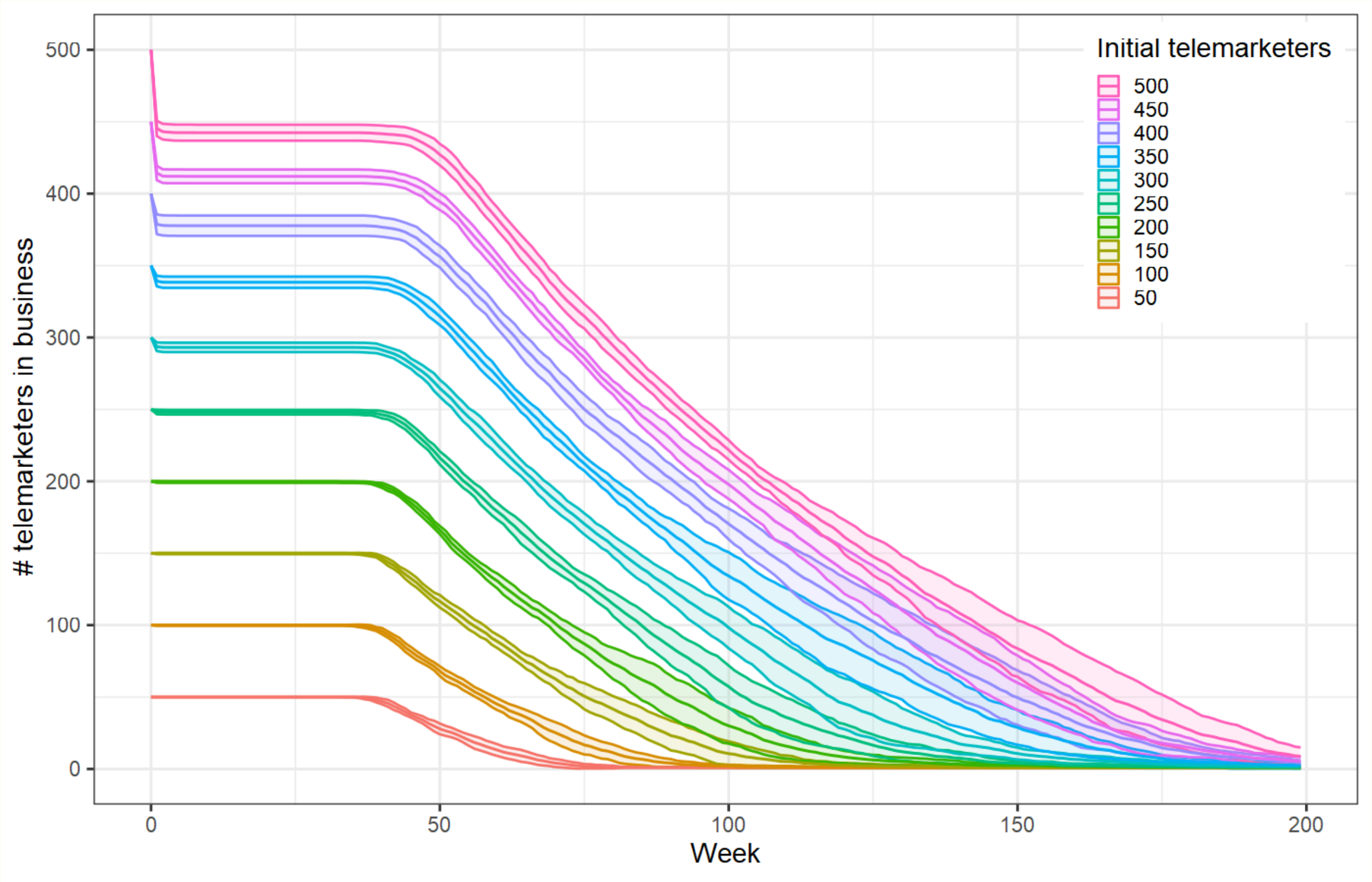
- Telemarketing firms interact
  - Telemarketer calls patches
  - If patch has received a previous call that tick, it hangs up
  - If patch has not received a previous call that tick, it buys something
  - Interaction is indirect, mediated by patches
- Accounting:
  - Net profit = 2 × sales − 50 × size
  - If balance < 0, firm goes bankrupt
- Growth
  - If balance > growth threshold, firm increases size proportional to excess balance

Results

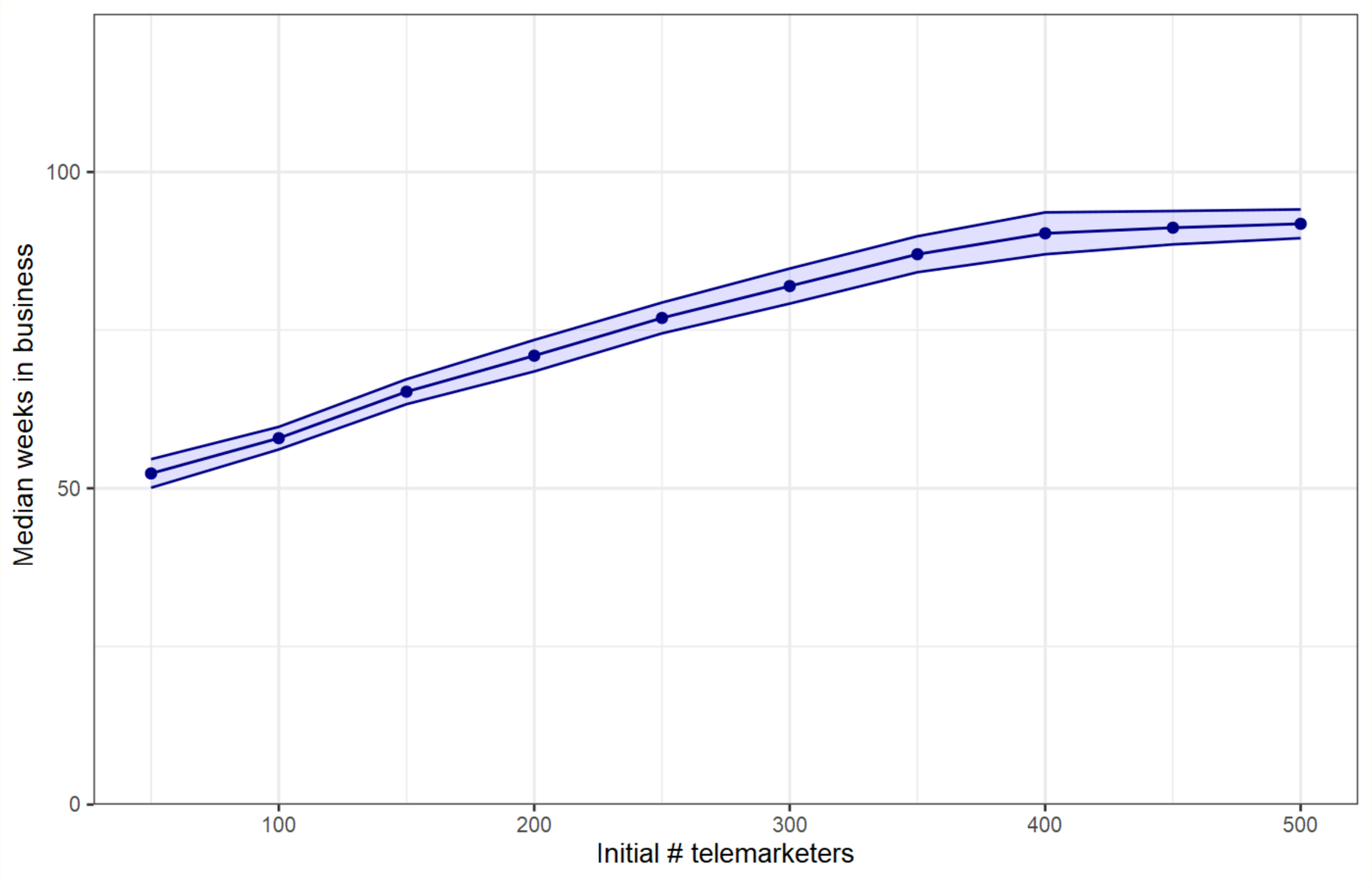
Results


Initial telemarketers
500
450
400
350
300
250
200
150
100
50
telemarketers in business
0
100
200
300
400
500
0
50
100
150
200
Week

Variation

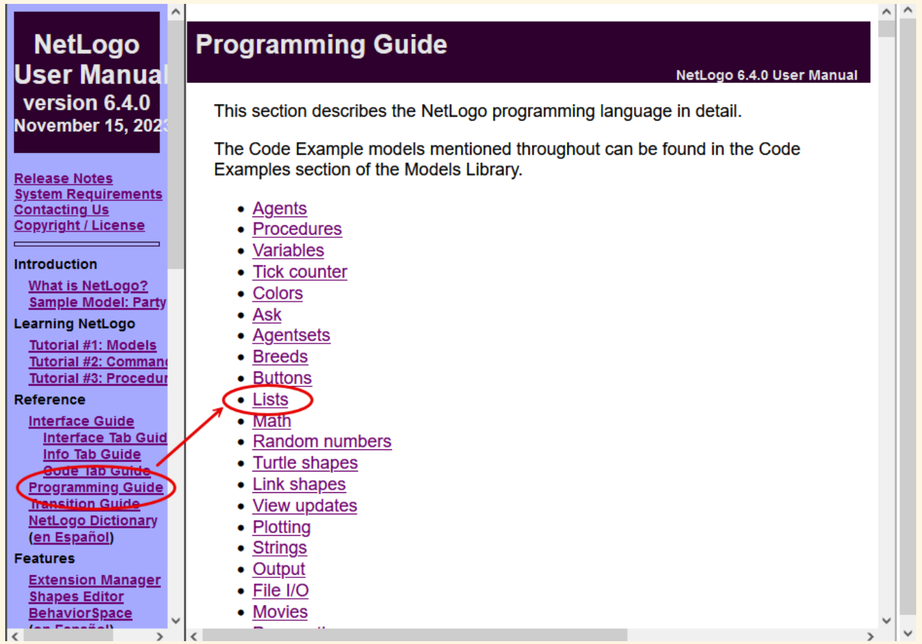

Initial telemarketers
500
450
400
350
300
250
200
150
100
50
telemarketers in business
500
400
300
200
100
0
0
50
100
150
200
Week

Median Weeks in Business



How to Calculate Median Weeks in Business?

- There are 50–500 telemarketers (turtles)
- What happens when a telemarketer goes out of business?
- What do you need, to calculate a median?
- We need a list of how long each turtle survives
- Use NetLogo `list` primitive in a global variable
- How do you figure out the age of a turtle in this model?
- How might we update the list? (Hint: consult the NetLogo dictionary)
  - You may want to use the `replace-item` primitive



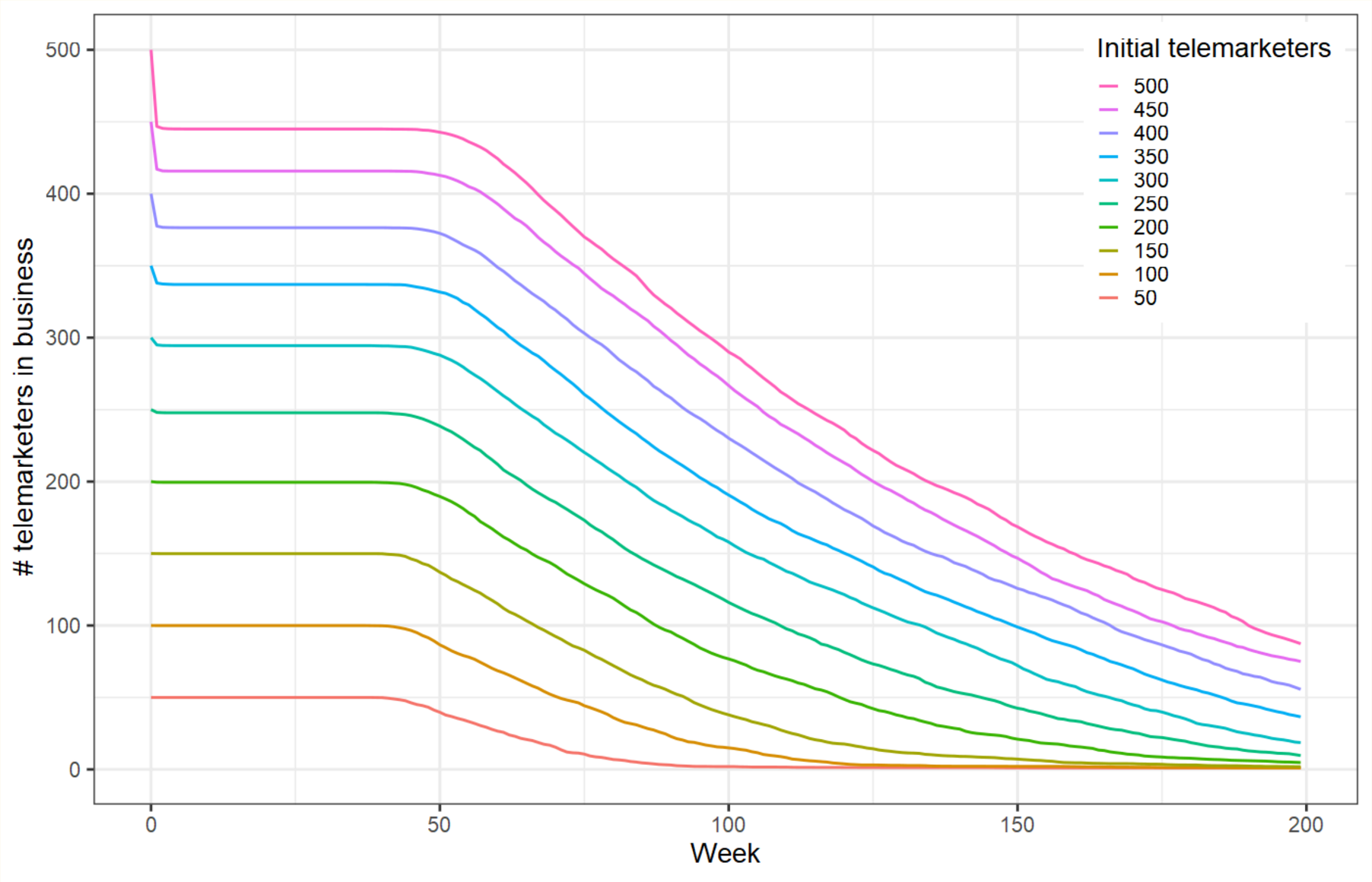
- See the “Lists” section in the NetLogo Programming Guide in the User Manual for more guidance on how to update lists.

Mergers

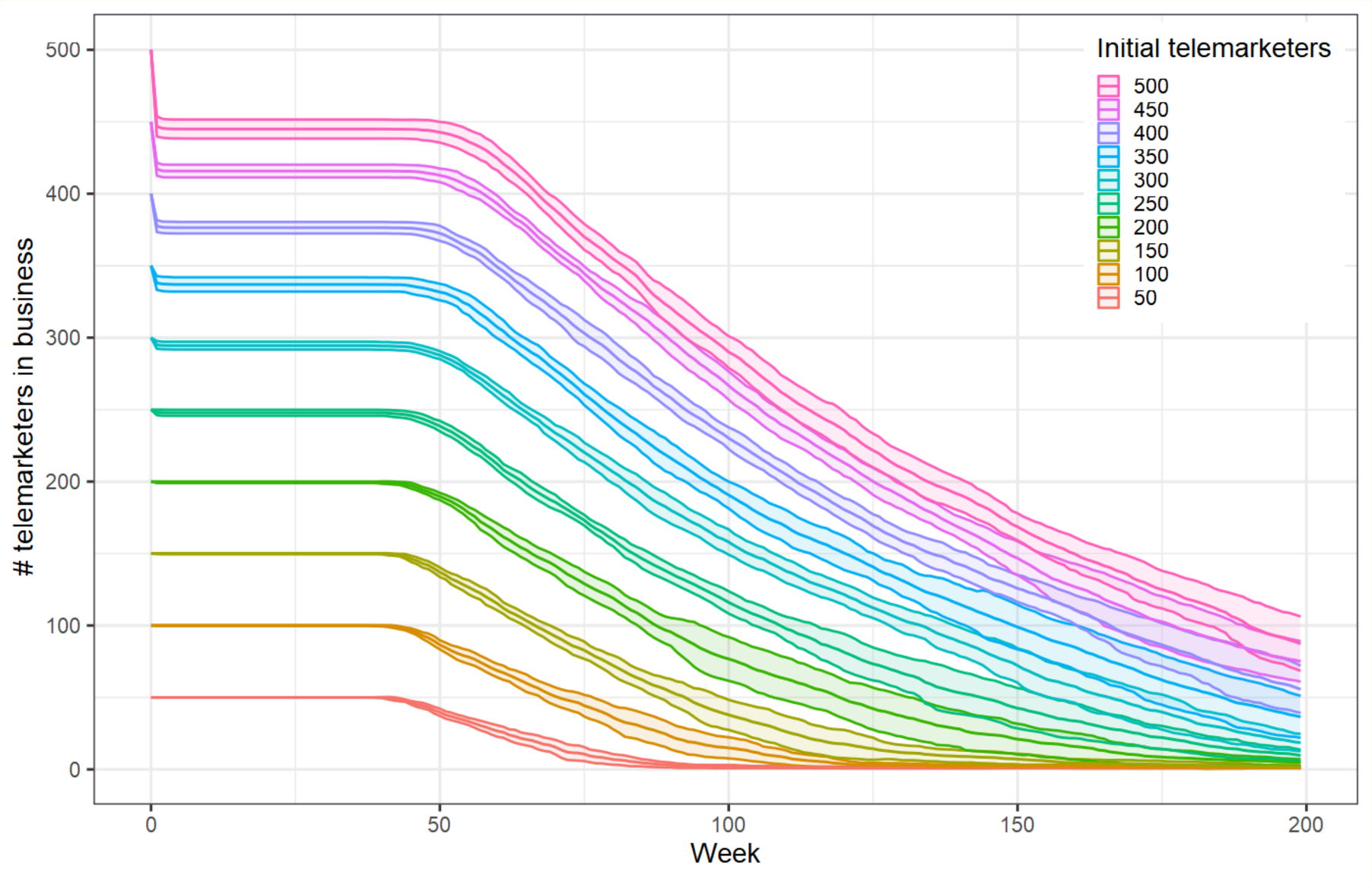
Mergers

- Instead of going bankrupt when the bank balance drops below 0, firms look for acquisition partner
  - Find a company that’s bigger and has enough money to pay off deficit.
  - If it finds a parent, parent pays off deficit (child firm ends up with 0 balance)
  - In future turns, child pays parent 50% of its net profits.
  - In future, if child’s balance becomes negative:
    - If parent has enough money, it pays child’s deficit
    - If parent does not have enough money, child dies.

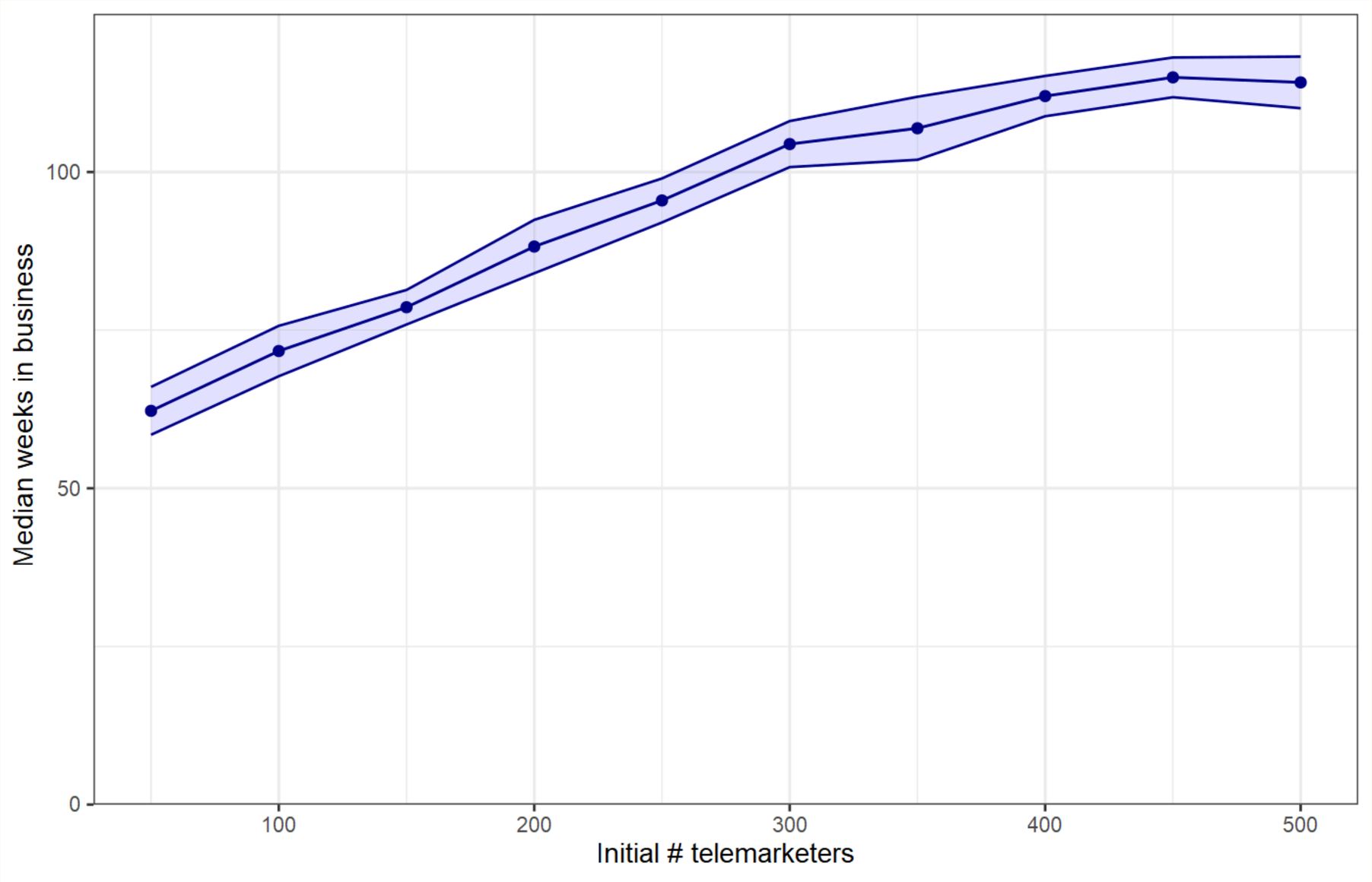
Results


Initial telemarketers
500
450
400
350
300
250
200
150
100
50
telemarketers in business
0
100
200
300
400
500
0
50
100
150
200
Week

Variation


Initial telemarketers
500
450
400
350
300
250
200
150
100
50
telemarketers in business
500
400
300
200
100
0
0
50
100
150
200
Week

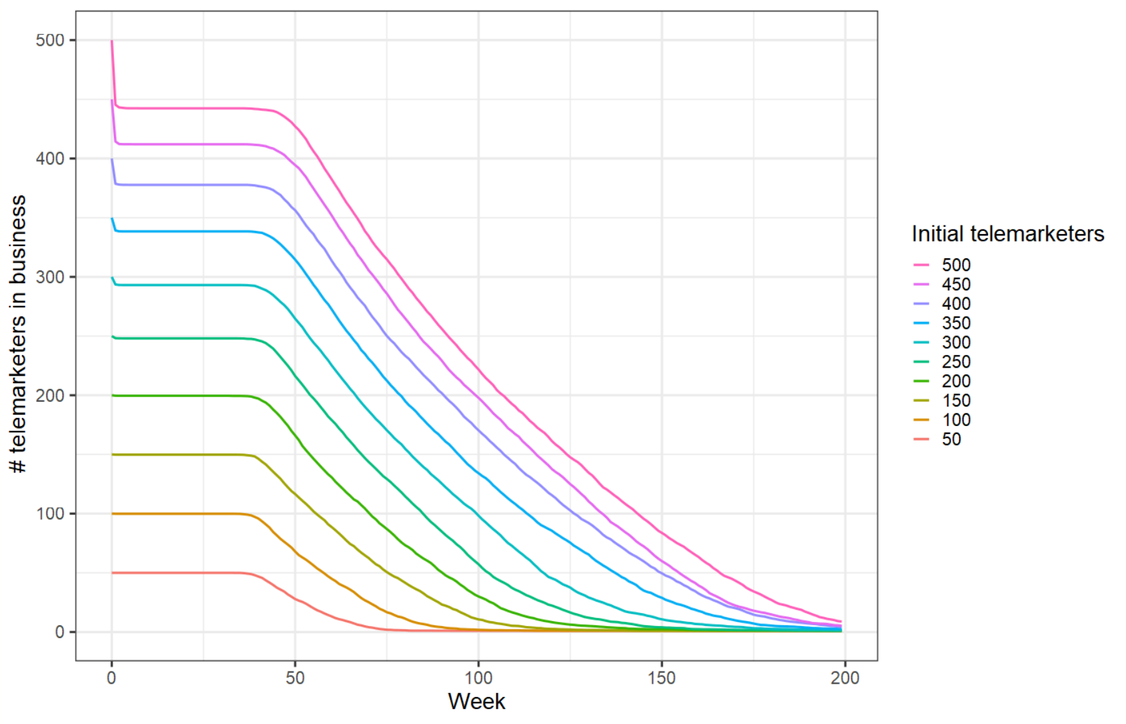
Median Weeks in Business



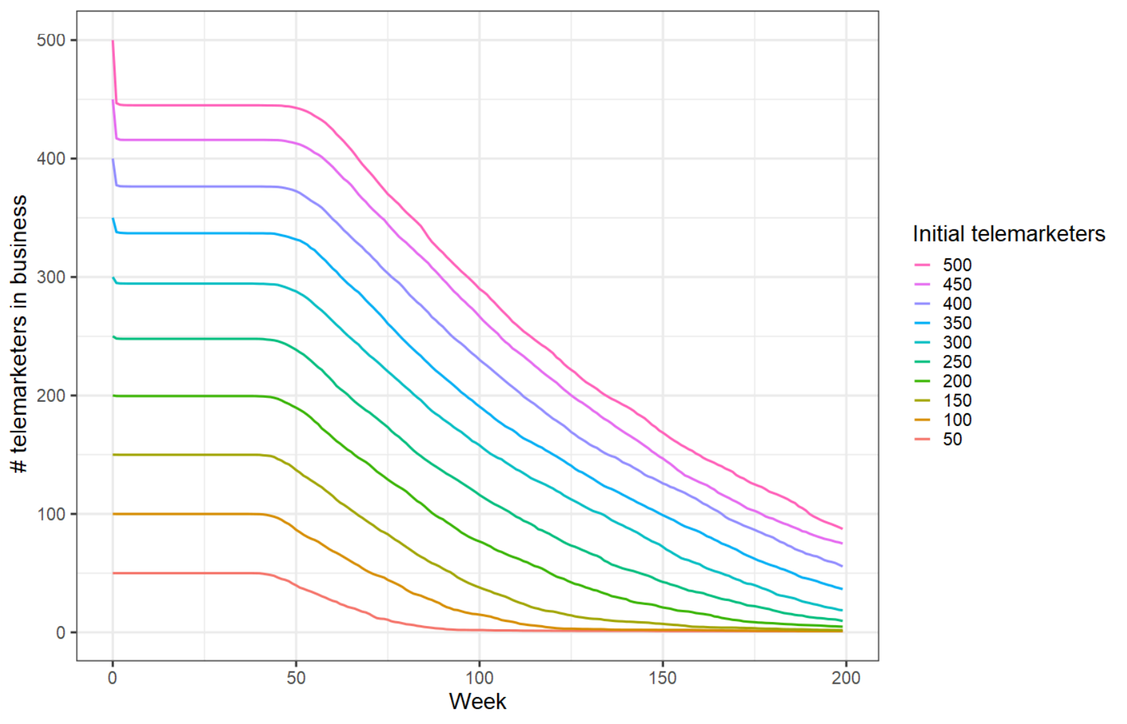
Comparisons

Progression

No Mergers

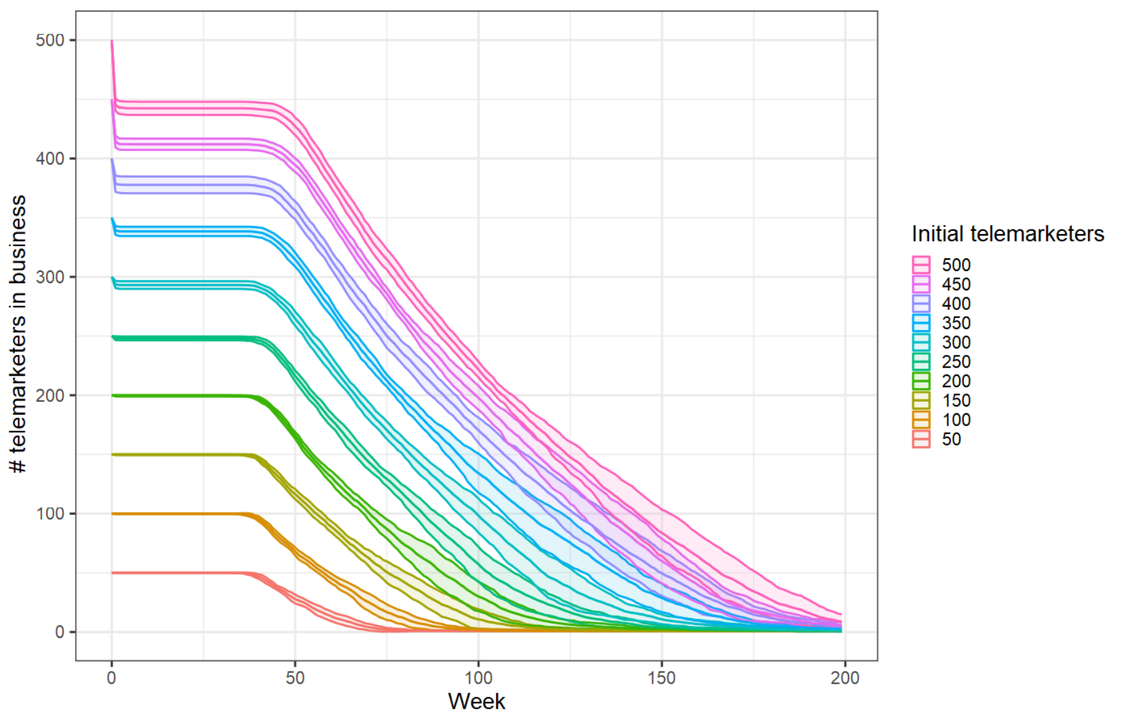


Mergers

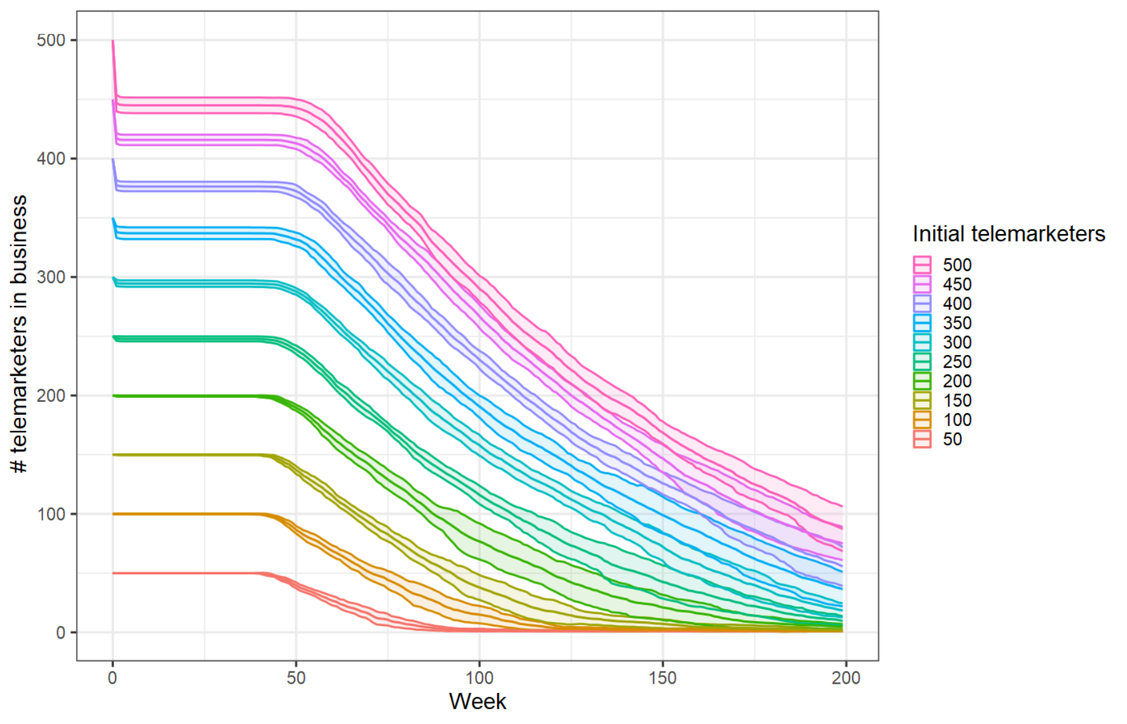


Variation

No Mergers

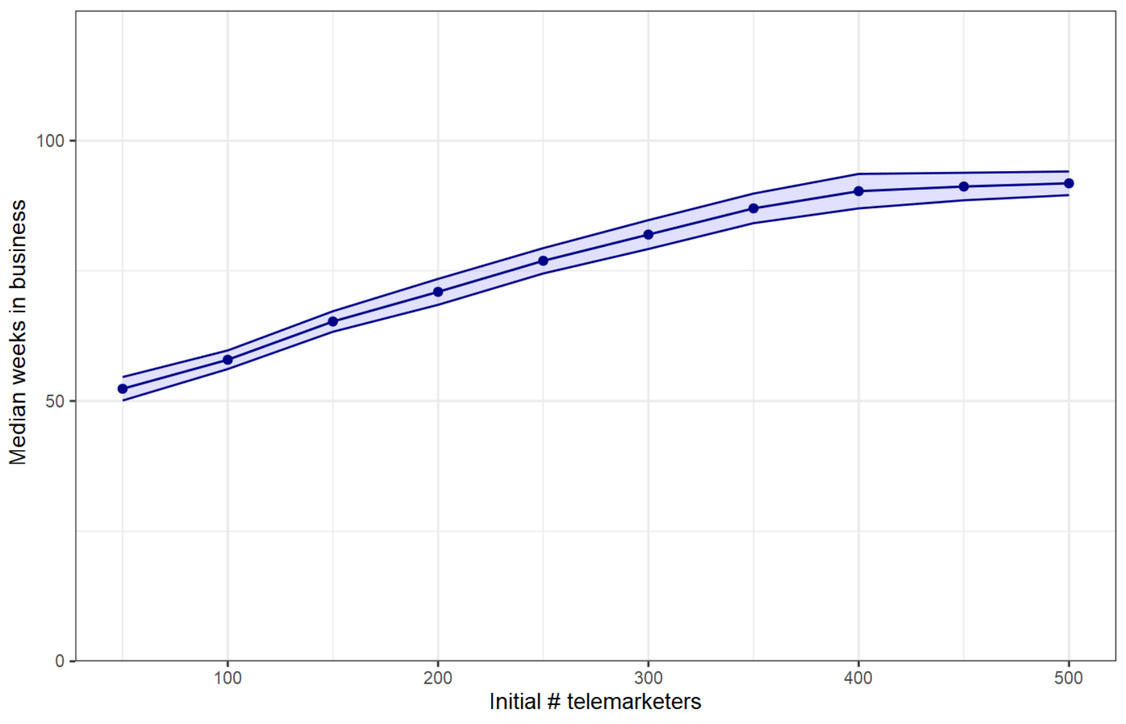


Mergers



Weeks in Business

No Mergers



Mergers

